Nitesh Nandan

Senior Software Engineer — Applied AI & Distributed Systems

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SUMMARY

Senior software engineer (8+ years) building production AI agents and high-scale distributed systems. Designed and shipped a supplier-facing **LangGraph** agent on **Google Vertex AI** that diagnoses product discoverability from live **BigQuery** data with multi-turn memory and layered guardrails, plus a team-wide AI tooling platform (**custom MCP server** + debug agent) that cut the hardest debugging task from hours to minutes. Backed by a deep systems foundation: **700M+ events/day** pipelines (scaling to **1.2B+**), a **~50% infra cost reduction**, and **P99 < 100ms** read APIs.

SKILLS

- **AI / LLM:** LangGraph, agentic state machines, A2A (Agent-to-Agent) protocol, MCP (Model Context Protocol), multi-turn memory / checkpointing, structured output, guardrails, prompt engineering, LLM observability (Arize)
- **Frameworks / Platforms:** LangChain, CrewAI, Google Vertex AI; model-agnostic LLM application development
- **Backend:** Java, Spring Boot, Python, FastAPI, gRPC, Kafka, event-driven pipelines
- **Data / Infra:** BigQuery, Bigtable, PostgreSQL, MySQL, Redis, Kubernetes, Docker, GCP, AWS
- **Observability:** Datadog, Arize, Grafana · **Quality:** SonarQube, pytest

WORK EXPERIENCE

Wayfair — Lead Software Engineer (L3) · 05/2024 – Present · Bengaluru

Findability domain: AI/agentic supplier tooling and high-RPS data-platform systems.

- **Designed and shipped the Findability Supplier Agent** — a production, supplier-facing conversational AI on a **10-node LangGraph state machine** (LLM on Google Vertex AI) that diagnoses product discoverability from **live BigQuery data**, with multi-turn **checkpointed memory**, **async parallel tool fan-out/fan-in**, and structured-output extraction (only 4 of 10 nodes call an LLM for cost and determinism).
- **Engineered deterministic guardrails enforced before the LLM** — accountability-based redaction, market scoping, and ownership/IDOR filtering — so suppliers never see internal barrier keys or another supplier's data; integrated into a multi-agent ecosystem over the **A2A protocol** with **MCP** tool federation and hot-reloadable config.
- **Built a team-wide AI productivity platform:** a custom **14-tool MCP server** giving read-only, cost-guardrailed access to BigQuery/PostgreSQL/Bigtable, and a **findability debug agent** that traces any identifier end-to-end across **5 systems and two data models** — cutting the domain's hardest debugging task from **3–4 hours to minutes** and turning a senior-only task into self-serve for the whole team.
- **Authored ~14 agentic SDLC workflow skills** (RFC → Jira → PR → review) plus a persistent domain-memory and team-convention model, standardizing how the team plans, reviews, and ships.
- **Scale & reliability:** operate an async event pipeline at **700M+ events/day** and authored the design proposal to re-architect it toward **1.2B+**; led a **zero-downtime CDF 2.0 migration** across PostgreSQL/BigQuery/Bigtable at **6–8K RPS peak** with zero consumer regression; sustained a high-traffic read API at **P99 < 100ms**.
- **Tech:** Python, LangGraph, Vertex AI, A2A, MCP, FastAPI, BigQuery, Bigtable, Java, Spring Boot, Kafka, Kubernetes, Arize, Datadog.

Meesho — Software Development Engineer III · 07/2022 – 05/2024 · Bengaluru

Backend owner on the Taxonomy and Supplier Store services (high-scale marketplace platform).

- Delivered **~50% infrastructure cost reduction** via L1/L2 caching, Redis pub/sub optimization, large-entity consolidation, and improved gRPC communication; cut DB cost a further **25%** through targeted schema and performance tuning.
- Re-architected the Supplier Store cache from scratch with senior architects, driving **Redis cost from $18/hr to $0.90/hr (~95% reduction)** and ~15 fewer pods; on Taxonomy, combined Redis + EKS optimizations saved **$70K+/month**; reduced Supplier Store **RDS cost by 50%** with no other cost regression.
- **Owned 5 production services** end to end, including **2 P0 systems sustaining 50K–70K RPS**.

- Took over and re-implemented the Hero Product API, now serving **~40K RPS**; built an independently-scaling Taxonomy Indexer pipeline that cut P1-flow alerts.
- Built a generic, config-driven **Kafka retry + dead-letter-queue** framework extensible across all consumers, making failed messages recoverable and eliminating silent message loss.
- **Tech:** Java, Spring Boot, Kafka, Redis, MySQL/RDS, gRPC, AWS EKS, Grafana, SonarQube.

Meesho — Software Development Engineer II · 06/2021 – 06/2022 · Bengaluru
Supplier-acquisition platform — onboarding, referrals, and supplier growth.

- Developed supplier-acquisition systems (Referrals, Supplier Data Platform, Agency Panel) supporting **4× growth in the supplier community within a year**, streamlining onboarding.
- Decoupled monolithic dependencies, improving service scalability and reliability.
- **Tech:** Java, Spring Boot, Kafka, MySQL, Redis.

imimobile (part of Cisco) **— Software Engineer** · 12/2019 – 06/2021 · Hyderabad

- Redesigned the group de-duplication workflow to cut group-creation time, building the supporting components with RabbitMQ, MySQL, PHP, and Docker.

HighRadius — Associate Software Engineer · 05/2018 – 06/2019 · Hyderabad

- Built an EDI (Electronic Data Interchange) parser from scratch, automating manual data exchange and cutting client-onboarding operational costs.

PERSONAL PROJECTS

- **Agent Memory MCP Server** *(Python, FastMCP, PostgreSQL, asyncpg, Docker)* — Built a Model Context Protocol server that gives LLM agents persistent memory (`put_memory` / `get_memory` / `get_memory_history`): atomically upserts the latest value **and** appends an immutable history log in a single transaction; project-scoped keys, stdio + SSE transports, containerized with auto-migrations.
- **LLM-Driven Algorithmic Trading Platform** *(Python, FastAPI, React, PostgreSQL, Redis, MCP)* — A personal end-to-end trading system:
  - **Agentic engine** — an MCP server exposing **18 tools** that uses an **LLM as the reasoning engine** to screen NSE equities (Wyckoff VSA strategy), confirm entries, place/modify Kite GTT orders, and autonomously exit positions on stop/target hits.
  - **Full-stack platform** — FastAPI + React app with multi-broker integration (Zerodha Kite, Upstox), 11 API modules, APScheduler-driven strategy execution, Redis-cached indicators, and a backtesting engine.
  - **Signal pipeline** — a 3-service architecture (research/backtest → daily screener → executor) communicating through a shared PostgreSQL contract with idempotent, fully-audited signal state transitions.
  - **Production deployment** — self-hosted on a Linux VPS: Nginx reverse proxy with TLS, Dockerized services on custom domains, and a shared PostgreSQL instance; infrastructure managed as code (Terraform on GCP).
- **Multi-DB Query Executor MCP** *(Python, FastMCP, asyncpg)* — Multi-project Postgres MCP server with per-project read-only enforcement (write/DDL blocked at the first SQL token), schema inspection, `EXPLAIN ANALYZE`, and slow-query analysis tools.
- **HTTP Web Server (from scratch)** — TCP connection handling, HTTP verb/header parsing, and concurrent connection management; a ground-up study of the protocol layer.

EDUCATION

B.Tech, Computer Science & Engineering — National Institute of Technology, Patna · *2014 – 2018*

ACHIEVEMENTS & CERTIFICATIONS

- HackerRank contests: 5 badges (1 Gold, 1 Silver); qualified for Google Code Jam Round 1; 100% on NIIT online DSA test.
- DeepLearning.AI: *Long-Term Agentic Memory with LangGraph*, *Multi-AI Agents & Advanced Use Cases with CrewAI* · Udemy: *MCP Masterclass*, *Java Multithreading & Concurrency*.